# Tilting

Exam code: 9MA0

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## Tilting

### What does it mean if a rod is on the point of tilting?

- A rod will **tilt** (rotate) about a **pivot** if the **resultant moment** about that pivot is **non-zero**
- Examples of tilting:
  - A person walking a plank on a ship will get to a certain distance before the plank tilts
  - Children playing on a seesaw

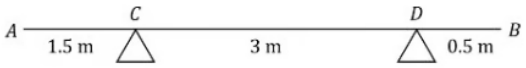
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

### How do I solve problems involving a rod on the point of tilting?

- **Taking moments about the pivot** is the key step as it allows you to ignore the reaction force
- The question might not use the phrase "tilting"
  - It could ask you how far along a plank that it is safe to walk (it is safe as long as the plank does not tilt)
  - It could ask you to find the maximum mass that could be hung from a rod that would keep it horizontal (as soon as it begins to tilt it is no longer horizontal)

## Worked Example

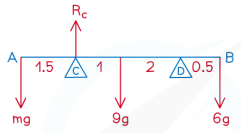
A uniform rod AB of mass 9 kg and length 5 m rests on two supports at two points C and D as shown in the diagram below.




A mass of 6 kg is placed at B and a mass of *m* kg is placed at A. Given that the rod is on the point of tilting about C, find the value of m.

**Answer:**

Point of tilting about C so reaction force at D is 0 N




Take moments about C so we can ignore $R_c$

Anti-clockwise Clockwise

M(C) $mg \times 1.5 = 9g \times 1 + 6g \times 3.5$

$1.5m\cancel{g} = 30\cancel{g}$ cancel the g's

$m = \frac{30}{1.5}$

$m = 20$

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## Examiner Tips and Tricks

- Remember to use common sense! If a rod was supported by two supports and a mass was placed on the rod in between these supports then the rod would never tilt (in real-life it might break though).

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